

Multi-Dimensional Tradeoffs and Mixing

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Motivating Example

- Suppose you are choosing between the following destinations:



Malaga



Alps

Motivating Example

- Suppose you are choosing between the following destinations:



Malaga



Alps

- Could be a hard choice if you care for BOTH mountains and beaches

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- Suppose your preferences are such that:
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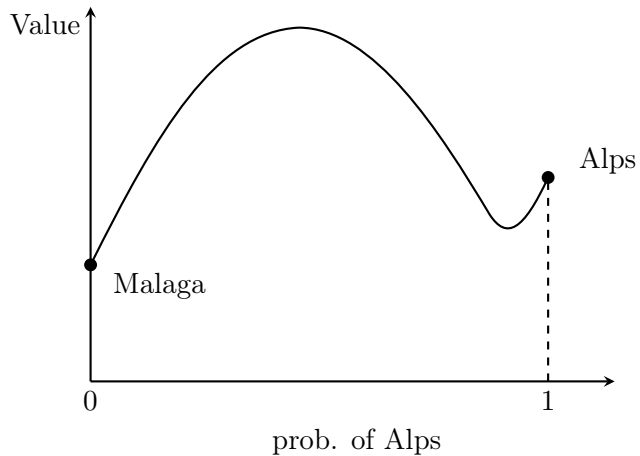
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- Suppose your preferences are such that:
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- Rationalizable by convex preferences. Suppose you also:
 - ▶ Alps $\succ$ 90-10 chance of Alps-Malaga

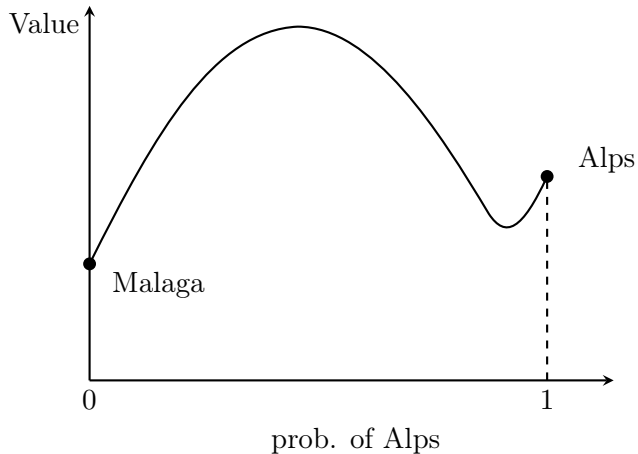
Motivating Example

- Suppose your preferences are such that:
 - ▶ $\text{Alps} \succ \text{Malaga}$
 - ▶ $50\text{-}50 \text{ chance of Alps-Malaga} \succ \text{Alps}$
- Rationalizable by convex preferences. Suppose you also:
 - ▶ $\text{Alps} \succ 90\text{-}10 \text{ chance of Alps-Malaga}$
- So you prefer the lottery if it is sufficiently random, but otherwise not.

Graphical Illustration

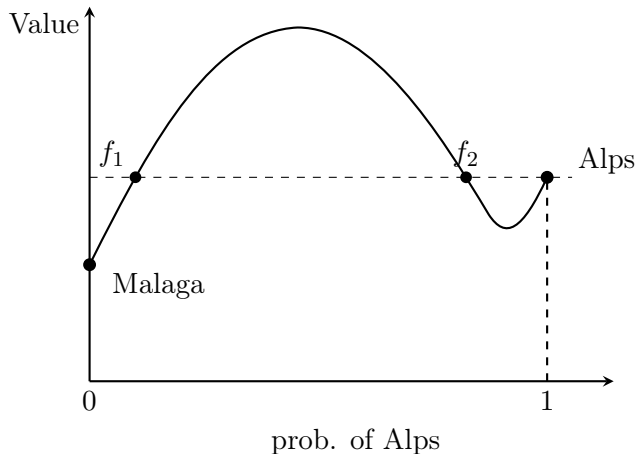


Graphical Illustration



- We call this choice behavior **Strong Non-Betweenness (SNB)**

Graphical Illustration



- SNB $\Rightarrow \exists$ two pairs of choices with different mixture attitudes

This Project

- Experimental evidence of Strong Non-Betweenness
 - ▶ Widespread and robust
 - ▶ High-stakes environment (choices between trips!)
- Characterize Strong Non-Betweenness
 - ▶ SNB equivalent to having mixture lovingness and aversion (ex: α -maxmin - but also more general models)
 - ▶ Equivalence is *structural* of multi-dimensional tradeoffs and not behavioral

Theory

Framework

- X, S - finite sets of outcomes and states
- $\mathcal{F}$ - set of Anscombe-Aumann acts
 - ▶ Notation: $f\alpha g = \alpha f + (1 - \alpha)g$

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Preference class $\mathcal{V}$

$\succsim \in \mathcal{V}$ satisfies completeness, transitivity, mixture continuity, statewise monotonicity, and strict FOSD on constant acts

Strong Non-Betweenness

Definition (Strong Non-Betweenness)

$\succsim$ displays Strong Non-Betweenness if $\exists f \sim g, \alpha, \beta \in (0, 1)$ such that

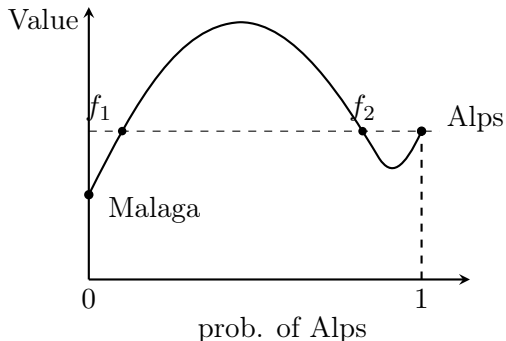
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Strong Non-Betweenness is not pathological

A deliberately standard benchmark

$$V(f) = \sum_{s \in S} \pi_s CE_s(f), \quad CE_s(f) = u^{-1}(\mathbb{E}_{f_s}[u(x)]) .$$

Within states

Expected utility over risk

Across states

Weighted sum of CEs

No exotic ingredient

Fixed probabilities,
additivity, separability

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Mixture attitude comes from how mixing changes tradeoffs across dimensions.

A Simple Example of Strong Non-Betweenness

$$u(x) = x^3, \quad CE_s(f) = (\mathbb{E}_{f_s}[x^3])^{1/3}, \quad V(f) = \frac{1}{2}CE_1(f) + \frac{1}{2}CE_2(f).$$

Two indifferent acts

$$f = (-1, 3), \quad g = (1, 1)$$

Since the acts are deterministic,

$$V(f) = V(g) = 1$$

But mixtures move both ways

$$V\left(\frac{1}{2}f + \frac{1}{2}g\right) = \frac{1}{2} \left(0 + \sqrt[3]{14}\right) \approx 1.205 > 1.$$

$$V(.9f + .1g) = \frac{1}{2} \left(\sqrt[3]{-.8} + \sqrt[3]{24.4}\right) \approx .986 < 1.$$

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Takeaway: Strong Non-Betweenness arises in a deliberately standard model

Strong Non-Betweenness and Mixture Non-Triviality

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- Immediate that Strong Non-Betweenness $\Rightarrow$ Mixture Non-Trivial
- For a large class of models, the opposite implication is unavoidable

Main Result

Theorem

The set of $\succsim$ in $\mathcal{V}$ where Strong Non-Betweenness $\Leftrightarrow$ Mixture Non-Trivial fails is meagre (and hence it holds in a dense set).

Topology

Define the binary lottery and constant binary act

$$\beta(t) := t\delta_{x_m} + (1 - t)\delta_{x_0}, \quad \bar{\beta}(t) := (\beta(t), \dots, \beta(t)).$$

Calibrated utility

For every $\succsim \in \mathcal{V}$, there is a unique continuous $U_{\succsim} : \mathcal{F} \rightarrow [0, 1]$ such that $f \sim \bar{\beta}(U_{\succsim}(f))$ and $U_{\succsim}(\bar{\beta}(t)) = t$

Interpretation: every act is measured by the constant binary act that makes the decision maker indifferent.

Topology Continued

The metric used in the theorem is

$$d_U(\zeta, \zeta') := \|U_\zeta - U_{\zeta'}\|_\infty = \sup_{f \in \mathcal{F}} |U_\zeta(f) - U_{\zeta'}(f)|.$$

Warm Up: $|X| = |S| = 2$

When $|X| = |S| = 2$, every act can be written as a point in a square:

$$f = (p_1, p_2) \in [0, 1]^2, \quad p_s = \Pr(x_1 \mid s).$$

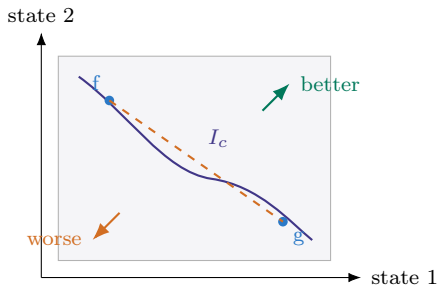
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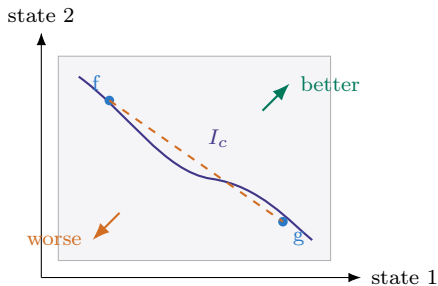


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In this case, MNT $\Leftrightarrow$ SNB everywhere

General Case

The general case has two steps:

- ① $\succsim$ violates equivalence $\Rightarrow \succsim$ is mixture independent somewhere
- ② The set of preferences that are mixture independent somewhere is meagre

Today, will only show that the set of preferences satisfying the equivalence is dense

Proof Sketch: Objects

Fix an interior level

$$I_c := \{f \in \mathcal{F} : U_{\tilde{z}}(f) = c\}, \quad c \in (0, 1).$$

For distinct $f, g \in I_c$, define the chord deviation

$$D_{f,g}(\lambda) := U_{\tilde{z}}(\lambda f + (1 - \lambda)g) - c, \quad \lambda \in [0, 1].$$

MNT

- $\exists f, g, f', g' \in I_c, \exists \lambda, \lambda'$ where
 $D_{f,g}(\lambda) > 0$ and $D_{f',g'}(\lambda') < 0$

SNB

- $\exists f, g \in I_c, \exists \lambda, \lambda'$ where
 $D_{f,g}(\lambda) > 0$ and $D_{f,g}(\lambda') < 0$

Geometry of Indifference Class

For each interior level I_c , study the indifferent pairs

$$C_2(I_c) := \{(f, g) \in I_c^2 : f \neq g\}.$$

Lemma

$C_2(I_c)$ is path-connected

Intuition.

- 1 Move every act inside I_c toward the calibrated constant act $\bar{\beta}(c)$.
- 2 Near $\bar{\beta}(c)$, the level has a local chart of dimension at least two.
- 3 Dim. $|S|(|X| - 1) \geq 2$ lets two points move around each other without collision.

Violating Equivalence $\Rightarrow$ Flat Chord

Define the average chord deviation

$$\Phi(f, g) := \int_0^1 D_{f,g}(\lambda) d\lambda, \quad (f, g) \in C_2(I_c).$$

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If I_c contains no SNB pair, every chord is single-signed. Further, if $\Phi = 0$, the chord must be flat:

$$U_{\tilde{z}}(\lambda f + (1 - \lambda)g) = c \quad \forall \lambda \in [0, 1].$$

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$$U_{\gtrsim}(\lambda f + (1 - \lambda)g) = c \quad \forall \lambda \in [0, 1].$$

Because $C_2(I_c)$ is path-connected and there is at least one positive and one negative chord by MNT, Φ must have a flat chord by IVT.

No Flat Chord Denseness

Lemma

For every $\zeta_0 \in \mathcal{V}$ and every $\varepsilon > 0$, there is $\zeta \in \mathcal{V}$ such that

$$d_U(\zeta, \zeta_0) < \varepsilon \quad \text{and} \quad U_\zeta \text{ has no flat chord.}$$

Fairly standard perturbation argument

Showing that $\mathcal{V}_{flat}$ is a countable union of nowhere dense set finishes

Interpretation

- Previous result shows the equivalence is common but might still wonder...
 - ▶ driven by particular psychological/behavioral channels that are dense?
 - ▶ maybe pathological models are dense rather than well-behaved models?
- Fundamentally: what is driving the equivalence?

Not Pathological (Again)

Proposition

Suppose $\succsim \in \mathcal{V}$ and satisfies Weak Certainty Independence, then

$$\text{Strong Non-Betweenness} \Leftrightarrow \text{Mixture Non-Trivial}$$

A few notes

- Equivalence is satisfied non-trivially for many models
- Implies the equivalence for the earlier sum of CEs model
- Weaker version of mixture non-triviality is enough

Structural

- Anscombe-Aumann chosen to illustrate the generality but...
- Theoretically: equivalence holds in other environments:
 - ▶ Lotteries over multi-dimensional vectors
 - ▶ Convex addition over multi-dimensional vectors
- Conceptually: Strong Non-Betw. arise in other interpretations of mixture
 - ▶ Dividing time between trip locations
 - ▶ Dividing budget between goods that are not always substitutes/complements
- Mixture attitudes can be driven by various channels, but the equivalence is structural and driven by multi-dimensional tradeoffs

Experiment

Experimental Questions

- We currently ask and answer two empirical questions:
 - ▶ Is Strong Non-Betweenness common?
 - ▶ Does the equivalence hold?
- Our analysis uses data from Toussaert (2024), will run additional experiment

Toussaert (2024) Set-Up (briefly)

- Elicit ranking over round trips to 10 destinations:
 - ▶ Both valuation and choice based - cross-checking consistency
 - ▶ High stakes - trips are all valued at £420

Choices and Strong Non-Betweenness

After ranking trips, $f_1 \succsim f_2 \dots \succsim f_{10}$, subjects choose between

- f_1 vs $(p, f_1; 1 - p, g)$
- $p \in \{0.9, 0.5, 0.1\}$
- $g \in \{f_2, f_3, f_5, f_{10}\}$

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Strong Non-Betweenness is implied if there $\exists p < q$ such that

$$pf_1 + (1 - p)g \succ f_1 \succ qf_1 + (1 - q)g$$

Result 1: Strong Betweenness is common

- 34/84 (41%) subjects display SNB for at least one comparison
- Happens mostly between f_1 and lotteries of (f_1, f_2) and (f_1, f_3)
 - ▶ f_1 vs (f_1, f_2) : 24%
 - ▶ f_2 vs (f_1, f_3) : 18%

Result 2: Strong Non-Betweenness and Mixture Non-Triviality

- Strong Non-Betweenness implies Mixture Non-Triviality:
 - ▶ Studying relationship requires independent measure of Mixture Non-Triviality

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- Strong Non-Betweenness implies Mixture Non-Triviality:
 - ▶ Studying relationship requires independent measure of Mixture Non-Triviality
- We consider a partial, weak measure

$$(p, f_1; 1 - p, g) \succ f_1 \quad \text{and} \quad g' \succ (q, f_1; 1 - q, g')$$

- ▶ As $f_1 \succ g, g'$, this implies both mixture aversion and lovingness
- ▶ As $g \approx g'$ can hold, weaker (easier to satisfy) than mixture non-triviality

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- 13/84 (16%) subjects satisfy the condition, only 1 fails SNB

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- 13/84 (16%) subjects satisfy the condition, only 1 fails SNB
- Classify subjects as MNT if condition holds OR display SNB

Table: Cross-tabulation of MNT and SNB

MNT	SNB		
	0	1	Total
0	57.8% (48)	0% (0)	57.8% (48)
1	1.2% (1)	41% (34)	42.2% (35)
Total	59.0% (49)	41.0% (34)	100% (83)

Conclusion

- Strong Non-Betweenness may appear as a strange behavior.
- This project:
 - ▶ Document Strong Non-Betweenness empirically
 - ▶ Show it is equivalent to mixture attitude being not globally one-sided
 - ▶ Equivalence is driven by multi-dimensional tradeoffs - not pathological/psychological channels